

Targeting Higher Credit Card Payments

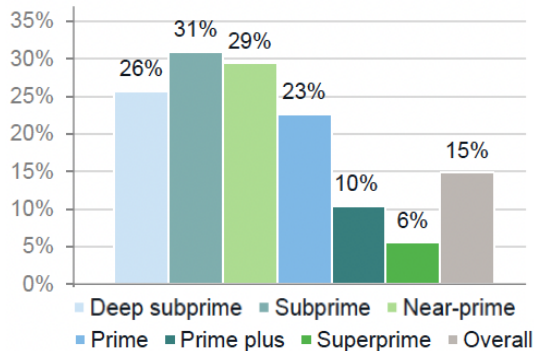
Benedict Guttman-Kenney

Rice University

Advances with Field Experiments, 17th September 2026

Credit Cardholders Often Only Pay The Minimum Payment Amount. Why?

% accounts pay the minimum, by credit score
(Consumer Financial Protection Bureau, 2025)



29% pay within \$50 of the minimum
(Keys & Wang, 2019 JFE)

One Potential Explanation: Anchoring Heuristic

Anchoring and adjustment heuristic: Foundational bias in Tversky & Kahneman (1974)

- Anchors: Salient but irrelevant numbers can bias choices
e.g., random digits of Social Security Number (Ariely et al., 2003)
- Consumers choose at / near the amount initially presented, adjust away too little

“One of the most robust phenomena studied across the social behavioral sciences”
– 50-year meta-analysis by Schley & Weingarten (2025)

Anchoring Influential In The Credit Card Literature

Research across the fields of economics, finance, management, marketing, and psychology attributes “**anchoring**” as an important determinant of consumer credit card payment choices

- Credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Thaler & Sunstein, 2008)

Anchoring Influential In The Credit Card Literature

Research across the fields of economics, finance, management, marketing, and psychology attributes “**anchoring**” as an important determinant of consumer credit card payment choices

- Credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Thaler & Sunstein, 2008)
- Shrouding minimum payment amount ↑ hypothetical credit card payments (Stewart, 2009)

Anchoring Influential In The Credit Card Literature

Research across the fields of economics, finance, management, marketing, and psychology attributes **“anchoring”** as an important determinant of consumer credit card payment choices

- Credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Thaler & Sunstein, 2008)
- Shrouding minimum payment amount ↑ hypothetical credit card payments (Stewart, 2009)
- Replicated across online experiments
(Navarro-Martinez et al., 2011; Salisbury, 2014; McHugh & Ranyard, 2016; Guttman-Kenney et al., 2018; Salisbury & Zhao, 2020; Hendy et al., 2021; Sakaguchi et al., 2022; Ortiz et al., 2024; Durán-Micco et al., 2026; Incekara-Hafalir et al., 2026)

Anchoring Influential In The Credit Card Literature

Research across the fields of economics, finance, management, marketing, and psychology attributes “**anchoring**” as an important determinant of consumer credit card payment choices

- Credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Thaler & Sunstein, 2008)
- Shrouding minimum payment amount ↑ hypothetical credit card payments (Stewart, 2009)
- Replicated across online experiments
(Navarro-Martinez et al., 2011; Salisbury, 2014; McHugh & Ranyard, 2016; Guttman-Kenney et al., 2018; Salisbury & Zhao, 2020; Hendy et al., 2021; Sakaguchi et al., 2022; Ortiz et al., 2024; Durán-Micco et al., 2026; Incekara-Hafalir et al., 2026)
- Some support in administrative data (Keys & Wang, 2019; Medina & Negrin, 2022) but does not explain autopay choices (Guttman-Kenney et al., 2025)

Anchoring Influential In The Credit Card Literature

Research across the fields of economics, finance, management, marketing, and psychology attributes “**anchoring**” as an important determinant of consumer credit card payment choices

- Credit card minimum payment “can serve as an anchor, and as a nudge that this minimum payment is an appropriate amount” (Thaler & Sunstein, 2008)
- Shrouding minimum payment amount ↑ hypothetical credit card payments (Stewart, 2009)
- Replicated across online experiments
(Navarro-Martinez et al., 2011; Salisbury, 2014; McHugh & Ranyard, 2016; Guttman-Kenney et al., 2018; Salisbury & Zhao, 2020; Hendy et al., 2021; Sakaguchi et al., 2022; Ortiz et al., 2024; Durán-Micco et al., 2026; Incekara-Hafalir et al., 2026)
- Some support in administrative data (Keys & Wang, 2019; Medina & Negrin, 2022) but does not explain autopay choices (Guttman-Kenney et al., 2025)

Key implication: Without anchoring, cardholders would ↑ credit card payments

This Paper: Do Online Experimental Findings Extrapolate To The Field?

Field experiment on 6,714 credit cardholders with a U.S. FinTech lender

- RCT that varies in-app (manual) credit card payment options for six statement cycles
 - Test shrouding minimum payment amount: Under anchoring, should $\uparrow$ payments
 - Test effects of 50% payment amount option
- Observe outcomes over six statement cycles
- Test predictions from conceptual framework for anchoring vs. targeting

This Paper: Do Online Experimental Findings Extrapolate To The Field?

Field experiment on 6,714 credit cardholders with a U.S. FinTech lender

- RCT that varies in-app (manual) credit card payment options for six statement cycles
 - Test shrouding minimum payment amount: Under anchoring, should $\uparrow$ payments
 - Test effects of 50% payment amount option
- Observe outcomes over six statement cycles
- Test predictions from conceptual framework for anchoring vs. targeting

Key Findings:

- Results of hypothetical online experiments do *not* translate into the field
- Evidence inconsistent with minimum payment acting as an anchor
 - No increase in payments from shrouding the minimum
 - Shrouding the minimum *lowers* initial payment choices

Contributions

- **Empirical methods:** Highly-replicated online experimental result does not hold up in field
(List, 2001, 2022; Gandhi et al., 2025)
- **Behavioral Economics:** Weak evidence of anchoring in markets suggests an opportunity to streamline behavioral biases
(Fudenberg, 2006; Stango & Zinman, 2023)
(Fudenberg et al., 2012; Maniadis et al., 2014; Alevy et al., 2015)
- Evidence that a **policy** shrouding the minimum payment would be ineffective
(Financial Conduct Authority, 2021; Brookings, 2022; Financial Health Network, 2023)
- **Household Finance:** Credit card minimum payment is a **target** rather than an anchor
– **Framework** for why payments respond to changes in the minimum but not nudges
(Keys & Wang, 2019; Seira et al., 2017; Adams et al., 2022; Medina & Negrin, 2022; Bartels et al., 2024; Guttman-Kenney et al., 2025; Katz et al., 2026; Schwartz, 2026)

1. The Experiment (Design, Data, Methodology)
2. Experimental Results

1. The Experiment

Field Experiment Design

Control (C)

9:41

< Choose Amount

👍

Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👍

Current Balance

\$74.00

Statement Balance

\$37.00

50% of Statement

\$11.00

Minimum Amount

Next

Field Experiment Design

Control (C)

9:41   

< Choose Amount

 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00  Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	\$11.00 Minimum Amount

Next

Treatment T1

9:41   

< Choose Amount

 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00  Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	

Next

Field Experiment Design

Control (C)

9:41   

< Choose Amount

 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00  Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	\$11.00 Minimum Amount

Next

Treatment T1

9:41   

< Choose Amount

 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00  Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	

Next

Treatment T2

9:41   

< Choose Amount

 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00  Current Balance	\$74.00 Statement Balance
--	-------------------------------------

Next

Field Experiment Design

Control (C)

9:41

Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👍 Current Balance **\$74.00** Statement Balance

\$37.00 50% of Statement **\$11.00** Minimum Amount

Next

Treatment T1

9:41

Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👍 Current Balance **\$74.00** Statement Balance

\$37.00 50% of Statement

Next

Treatment T2

9:41

Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👍 Current Balance **\$74.00** Statement Balance

Next

Prompt

9:41

Choose Amount

! Paying less than your minimum amount \$11.00 will likely result in interest and late fee charges.

\$5.00 ×

Your statement balance is due on April 24

\$124.00 👍 Current Balance **\$74.00** Statement Balance

Next

1 2 3
4 5 6
7 8 9
0 ×

If attempt to pay < minimum, a **prompt** reveals minimum payment

Field Experiment Design ($N = 6,714$)

Control (C)

4,542

9:41

< Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👍 Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	\$11.00 Minimum Amount

Next

Treatment T1

1,096

9:41

< Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👍 Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	

Next

Treatment T2

1,076

9:41

< Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 👍 Current Balance	\$74.00 Statement Balance

Next

T1 (vs C): Effect of *shrouding* minimum payment

T2 (vs C): Effect of *shrouding* minimum payment & 50% statement balance options

Field Experiment Design ($N = 6,714$)

Control (C)

4,542

9:41

< Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 🙌 Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	\$11.00 Minimum Amount

Next

Treatment T1

1,096

9:41

< Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 🙌 Current Balance	\$74.00 Statement Balance
\$37.00 50% of Statement	

Next

Treatment T2

1,076

9:41

< Choose Amount

👍 Paying your current balance avoids interest, boosts your credit score, and lowers utilization.

\$124.00

Your statement balance is due on April 24

\$124.00 🙌 Current Balance	\$74.00 Statement Balance

Next

T1 (vs C): Effect of *shrouding* minimum payment

T2 (vs C): Effect of *shrouding* minimum payment & 50% statement balance options

T1 (vs T2): Effect of *adding* 50% option (conditional on shrouding minimum)

Arro: U.S. FinTech credit card lender:

- Means: \$290 credit limits, \$200 statement balance, 560 VantageScore
- Clearly not representative of broader credit card market!

Below-prime credit score (< 660 VantageScore) consumers:

- Highest interest rates $\rightarrow$ largest potential gains from increasing payments
(Drechsler et al., 2025; Consumer Financial Protection Bureau, 2025)
- Most likely to pay at or just-above the minimum; similar “anchoring” estimates to prime
(Keys & Wang, 2019)

Arro: U.S. FinTech credit card lender:

- Means: \$290 credit limits, \$200 statement balance, 560 VantageScore
- Clearly not representative of broader credit card market!

Below-prime credit score (< 660 VantageScore) consumers:

- Highest interest rates $\rightarrow$ largest potential gains from increasing payments
(Drechsler et al., 2025; Consumer Financial Protection Bureau, 2025)
- Most likely to pay at or just-above the minimum; similar “anchoring” estimates to prime
(Keys & Wang, 2019)

Datasets

Lender administrative data linked with: (1) Clickstream data on payment choices;
(2) Plaid financial transaction data on liquid cash; (3) Equifax credit reporting data

Empirical Methodology & Predictions from Conceptual Framework

Dynamic regression specification:

$$Y_{i,t} = \sum_{\tau=0}^6 \left[\delta_{\tau}^{T1(vsC)} (D_{\tau} \times T1_i) + \delta_{\tau}^{T2(vsC)} (D_{\tau} \times T2_i) \right] + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (1)$$

- X_i vector of pre-experiment controls, μ_t statement cycle fixed effects
- Standard errors clustered at consumer-level

Empirical Methodology & Predictions from Conceptual Framework

Dynamic regression specification:

$$Y_{i,t} = \sum_{\tau=0}^6 \left[\delta_{\tau}^{T1(vsC)} (D_{\tau} \times T1_i) + \delta_{\tau}^{T2(vsC)} (D_{\tau} \times T2_i) \right] + X_i' \beta + \mu_t + \varepsilon_{i,t} \quad (1)$$

- X_i vector of pre-experiment controls, μ_t statement cycle fixed effects
- Standard errors clustered at consumer-level

Conceptual Framework Predictions

Under anchoring:

- Shrouding minimum $\uparrow$ initial payments, $\uparrow$ actual payments, & shifts distribution

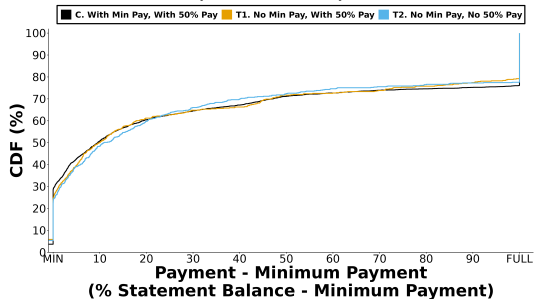
Under targeting:

- Shrouding minimum $\downarrow$ initial payments (reveal minimum), & no change in actual payments

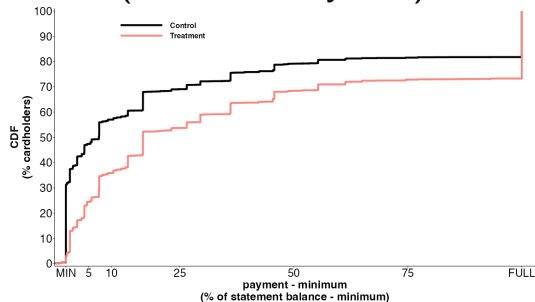
2. Experimental Results

Payment Distribution: Field Experiment Vs. Online Experiments

Field Experiment (This Paper)



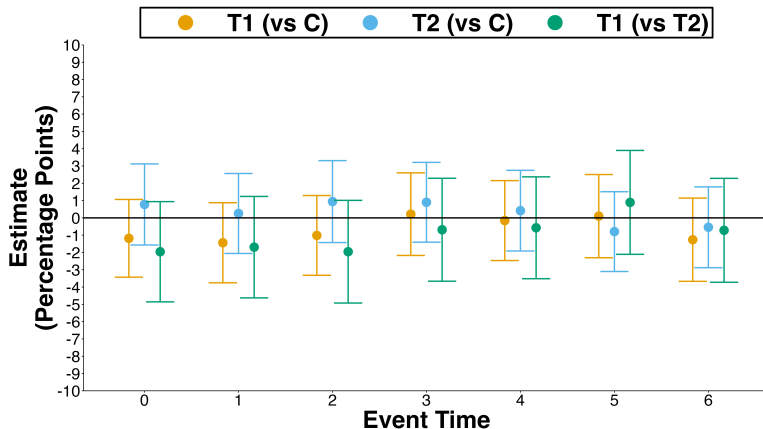
Online Experiment Example (Guttman-Kenney, 2024)



↑ 12.4*** p.p. (s.e. 0.8)

$N = 7,938$, mean 28.1%

Null Effects On Primary Outcome: Payments (% Statement Balance)



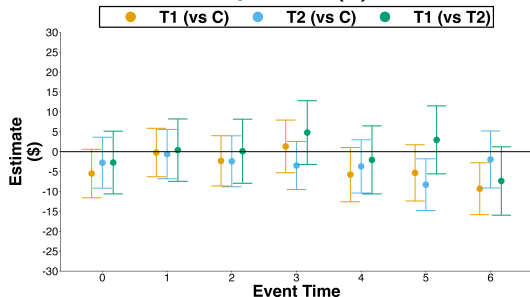
δ_6^k (percentage points, C mean: 27.8%):

T1 (vs C) -1.26 (s.e. 1.23) **T2 (vs C)** -0.54 (s.e. 1.19) **T1 (vs T2)** -0.72 (s.e. 1.53)

Pooling across cycles increases precision, does not change results

No Significant $\uparrow$ Payments (\$) Or $\downarrow$ Revolving Debt (\$)

A. Payments (\$)



T1 (vs C) —\$9.31 (s.e. \$3.32)**

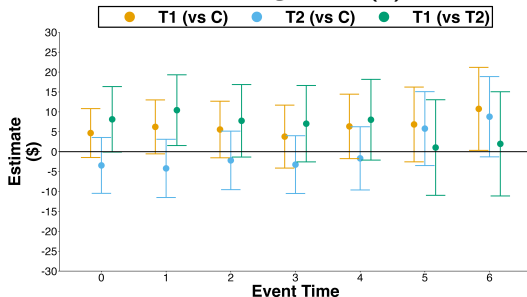
T2 (vs C) —\$1.95 (s.e. \$3.65)

T1 (vs T2) —\$7.36 (s.e. \$4.38)

C mean: \$58.4

The 50% option temporarily makes consumers less likely to pay in full

B. Revolving Debt (\$)



T1 (vs C): \$10.76* (s.e. \$5.32)

T2 (vs C): \$8.79 (s.e. \$5.14)

T1 (vs T2): \$1.97 (s.e. \$6.68)

C mean: \$230.9

δ_6^k (\$):

Clickstream Data Shows How Consumers “Hack” The Nudge

- Choose Low Initial Payment Amounts To Reveal Minimum, Then Pay It

Pr (Initial Payment Choice < Minimum Payment):

T1 (vs C) 19.44*** (s.e. 1.16) **T2 (vs C)** 30.50*** (s.e. 1.02)

(percentage points, C mean: 33.7%)

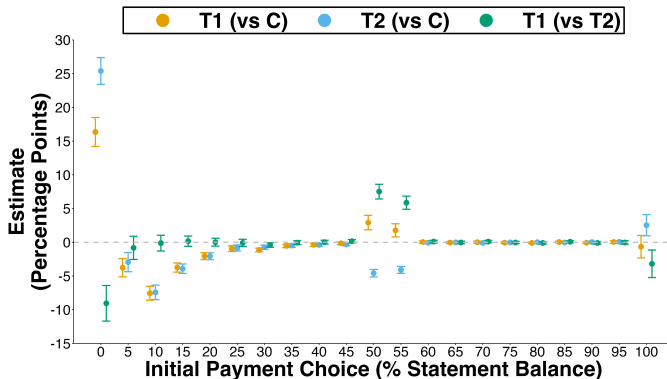
Clickstream Data Shows How Consumers “Hack” The Nudge

– Choose Low Initial Payment Amounts To Reveal Minimum, Then Pay It

Pr (Initial Payment Choice < Minimum Payment):

T1 (vs C) 19.44*** (s.e. 1.16) T2 (vs C) 30.50*** (s.e. 1.02)

(percentage points, C mean: 33.7%)



Cardholders Not At Borrowing Constraints But Frequently Lack Liquid Cash

1. **Credit Card Portfolio Utilization (Equifax):** 25% (median), 36% (mean)
 - Only 16% of cardholders with 90%+ portfolio utilization
 - Typically not at borrowing constraint (inconsistent with liquidity constraints in HANK)
 - Not significant effects for low-utilization consumers

Cardholders Not At Borrowing Constraints But Frequently Lack Liquid Cash

1. **Credit Card Portfolio Utilization (Equifax):** 25% (median), 36% (mean)
 - Only 16% of cardholders with 90%+ portfolio utilization
 - Typically not at borrowing constraint (inconsistent with liquidity constraints in HANK)
 - Not significant effects for low-utilization consumers
2. **Liquid Cash (Plaid):** *Minimum* liquid cash balances in 90 days pre-experiment
 - 63% of cardholders have <\$1, 93% <\$501, 96% <\$1,001 minimum liquid cash balances
 - Not significant heterogeneous effects by minimum liquid cash balances, however, cannot rule out meaningful positive/negative effects for \$1,001+ group ($N = 292$)
 - Similar pattern for large U.K. bank credit cardholders (Guttman-Kenney et al., 2025)

Cardholders Not At Borrowing Constraints But Frequently Lack Liquid Cash

1. **Credit Card Portfolio Utilization (Equifax):** 25% (median), 36% (mean)
 - Only 16% of cardholders with 90%+ portfolio utilization
 - Typically not at borrowing constraint (inconsistent with liquidity constraints in HANK)
 - Not significant effects for low-utilization consumers
2. **Liquid Cash (Plaid):** *Minimum* liquid cash balances in 90 days pre-experiment
 - 63% of cardholders have <\$1, 93% <\$501, 96% <\$1,001 minimum liquid cash balances
 - Not significant heterogeneous effects by minimum liquid cash balances, however, cannot rule out meaningful positive/negative effects for \$1,001+ group ($N = 292$)
 - Similar pattern for large U.K. bank credit cardholders (Guttman-Kenney et al., 2025)

My results support the macroeconomic modeling approach of Lee & Maxted (2026)

- present bias jointly explains high MPCs & revolving modest amounts of credit card debt (Meier & Sprenger, 2010; Kuchler & Pagel, 2021; Laibson et al., 2026)

Why Anchoring Appears in Hypotheticals But Not in the Field:

- Model anchoring as psychological cost of straying from displayed number ($\kappa_A/(\kappa_A + |U''|)$)
 - In online experiments, $\approx$ flat marginal value of cash $\rightarrow$ anchoring
 - In the field, steep marginal value of cash $\rightarrow$ no anchoring

Implications of Targeting from Conceptual Framework

Why Anchoring Appears in Hypotheticals But Not in the Field:

- Model anchoring as psychological cost of straying from displayed number ($\kappa_A / (\kappa_A + |U''|)$)
 - In online experiments, $\approx$ flat marginal value of cash $\rightarrow$ anchoring
 - In the field, steep marginal value of cash $\rightarrow$ no anchoring

Changing Minimum (Not its Display) Moves Payments:

- Nudging intermediate amounts cannot compete with payoff notches (minimum; full) (Agarwal et al., 2015; Keys & Wang, 2016; Seira et al., 2017; Adams et al., 2022)
- Increasing minimum payments reduces debt of unconstrained cardholders via 3 channels: (Keys & Wang, 2019; Medina & Negrin, 2022; Castellanos et al., 2026; Allen et al., 2026)
 1. Consumers bunching at/near-minimum move with minimum
 2. Higher minimum shrinks distance to full, so some jump to pay in full (Schwartz, 2026)
 3. Indexed targets (e.g., 2x minimum) follow higher minimum (Medina & Negrin, 2022)

- Demonstrate a highly-replicated online experimental result does not extrapolate to the field
- Consumers initially choose amount $<$ minimum to reveal it, and then pay it
- The minimum payment is a notch in economic payoffs.
It acts as a (relevant) target **not** an (irrelevant) anchor

Thank you!



www.benedictgk.com



benedictgk@rice.edu

Conceptual Framework: The Consumer's Problem

A consumer chooses payment $p \in [0, B]$ to maximize their utility:

$$U(p) = \underbrace{-C(p; a)}_{\text{cash cost}} + \underbrace{\rho p}_{\text{future benefit}} - \underbrace{F \cdot \mathbf{1}\{p < m\}}_{\text{fee}} + \underbrace{\eta_m \mu(p - m) + \eta_B \mu(p - B)}_{\text{reference dependence}} \quad (4)$$

$$p^*(a) = \begin{cases} B & a \geq B + x_H \\ a - x_H & m + x_H < a < B + x_H \\ m & m + x_F \leq a \leq m + x_H \\ \max\{a - x_L, 0\} & a < m + x_F \end{cases} \quad (5)$$

- x_H cash buffer once minimum is met
- x_L smaller cash buffer when fee unavoidable ($x_L \leq x_H$)
- x_F smallest cash buffer to avoid fee ($x_F < x_L$)

Contribution vs. Guttman-Kenney, Adams, Laibson, Stewart, & Leary (2025)

Results are consistent, despite different treatments and populations (large UK bank vs. small US FinTech)

1. Shroud minimum in manual payment vs. shrouding autopay minimum at card opening
 - Even without offsetting responses found in my earlier paper (autopay enrollment, offsetting higher autopay with lower manual payments), shrouding remains ineffective
2. Test a new intermediate payment option: 50% of the statement balance
3. **Most importantly, the framework and merged data enable me to explain:**
 - (i) **why consumers pay at or just above the minimum**
 - (ii) **why behavior moves with the minimum but not with nudges**

Pre-Experiment Controls

Linear controls:

- Age
- Credit Score
- Card Tenure
- Pre-experiment values of all of the primary and secondary outcomes
- Number of open credit cards held in their portfolio
- Total credit card portfolio balances
- Total credit card portfolio credit limits

Fixed effects:

- Interest rate
- Credit limit
- Autopay type

Primary and Secondary Outcomes

Primary Outcome: 1. Payments (%) = $\sum \text{Payments} / \text{Statement Balance}$

Secondary Outcomes:

2. Payments (\$)
3. Statement Balance (\$)
4. Spending (\$)
5. Spending (%)
6. Revolving Debt (\$)
7. Credit Limit (\$)
8. Miss Pay (%)
9. Pay Min (%)
10. Pay 50% (%)
11. Pay Full (%)
12. Pay Current Balance (%)

Attrition (3, 6 Cycles)

	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
3	-0.23 (0.14)	-0.12 (0.18)	-0.12 (0.18)	0.37%
6	-0.09 (0.64)	0.00 (0.70)	-0.09 (0.83)	5.00%
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Pre-Experiment Summary Statistics

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 – C: Mean (4)	[p-value] (5)	T2 – C: Mean (6)	[p-value] (7)
Age	45.51	45.68	45.84	0.17 (0.38)	[0.652]	0.33 (0.39)	[0.390]
Credit Score	561.70	561.06	562.10	–0.64 (2.06)	[0.757]	0.40 (2.10)	[0.847]
APR (%)	15.71	15.72	15.72	0 (0.01)	[0.826]	0 (0.01)	[0.850]
Tenure (days)	265.58	257.53	262.24	–8.05 (5.10)	[0.114]	–3.34 (5.10)	[0.512]
Autopayer (%)	41.19	43.59	39.69	2.39 (1.68)	[0.154]	–1.50 (1.65)	[0.362]
Payments (%)	45.06	48.00	44.34	2.95 (1.44)	[0.040]	–0.72 (1.39)	[0.608]
Payments	77.07	85.00	76.99	7.93 (4.21)	[0.060]	–0.08 (3.79)	[0.984]
Statement Balance	207.94	206.38	208.75	–1.55 (7.57)	[0.837]	0.82 (7.41)	[0.912]
Spending	98.13	99.81	101.30	1.68 (4.67)	[0.719]	3.17 (4.82)	[0.510]
Spending (%)	50.11	51.95	50.25	1.84 (1.46)	[0.207]	0.14 (1.44)	[0.925]
Revolving Debt	149.09	144.62	149.98	–4.47 (7.15)	[0.532]	0.89 (6.98)	[0.898]
Credit Limit	292.09	287.64	294.11	–4.45 (8.16)	[0.585]	2.02 (8.13)	[0.803]
Miss Pay (%)	17.00	14.59	15.88	–2.41 (1.21)	[0.047]	–1.12 (1.24)	[0.365]
Pay Min (%)	17.48	17.38	16.79	–0.10 (1.29)	[0.937]	–0.69 (1.26)	[0.583]
Pay 50% (%)	3.74	3.07	4.20	–0.68 (0.60)	[0.257]	0.45 (0.67)	[0.497]
Pay Full (%)	31.73	34.39	30.02	2.66 (1.61)	[0.098]	–1.71 (1.55)	[0.270]
Pay Current Balance (%)	9.25	11.15	9.67	1.91 (1.05)	[0.070]	0.42 (0.99)	[0.668]
Minimum Payment	14.92	15.08	15.43	0.16 (0.4)	[0.692]	0.51 (0.41)	[0.216]

T-Tests (3 Cycles)

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 – C (4)	T2 – C (5)	T1 – T2 (6)
Payments (%)	32.92	34.40	33.83	1.48 (1.39) [0.285]	0.92 (1.37) [0.503]	0.57 (1.76) [0.747]
Payments	59.01	63.18	55.75	4.16 (4.00) [0.297]	-3.26 (3.61) [0.367]	7.42 (4.81) [0.123]
Statement Balance	231.01	233.85	225.52	2.84 (7.93) [0.720]	-5.49 (7.65) [0.473]	8.33 (9.89) [0.400]
Spending	71.94	73.64	67.50	1.70 (4.54) [0.708]	-4.43 (4.27) [0.299]	6.14 (5.57) [0.271]
Spending (%)	36.03	38.04	36.55	2.01 (1.43) [0.158]	0.52 (1.42) [0.714]	1.49 (1.82) [0.412]
Revolving Debt	187.21	187.88	182.58	0.67 (7.70) [0.931]	-4.63 (7.48) [0.536]	5.30 (9.65) [0.583]
Credit Limit	320.21	319.02	321.35	-1.20 (8.75) [0.891]	1.14 (8.64) [0.895]	-2.34 (11.02) [0.832]
Miss Pay (%)	28.82	29.58	29.09	0.76 (1.55) [0.622]	0.28 (1.53) [0.857]	0.49 (1.96) [0.803]
Pay Min (%)	16.20	11.63	11.62	-4.57*** (1.12) [0.000]	-4.58*** (1.11) [0.000]	0.01 (1.38) [0.995]
Pay 50% (%)	2.72	3.35	0	0.63 (0.60) [0.293]	-2.72*** (0.24) [0.000]	3.35*** (0.55) [0.000]
Pay Full (%)	22.76	24.19	24.25	1.42 (1.45) [0.326]	1.48 (1.44) [0.303]	-0.06 (1.84) [0.974]
Pay Current Balance (%)	5.90	5.86	6.50	-0.04 (0.80) [0.960]	0.60 (0.82) [0.470]	-0.64 (1.03) [0.539]

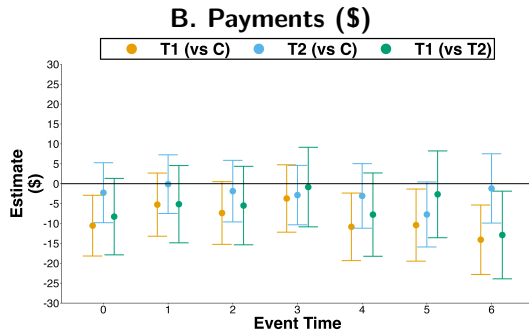
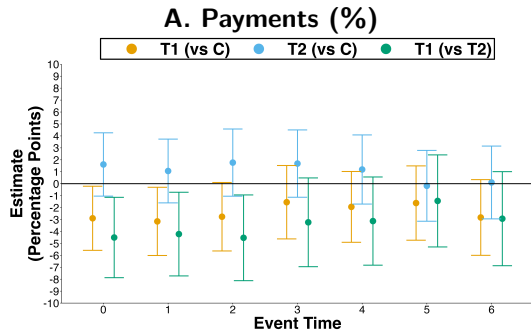
T-Tests (6 Cycles)

Outcome	C Mean (1)	T1 Mean (2)	T2 Mean (3)	T1 - C (4)	T2 - C (5)	T1 - T2 (6)
Payments (%)	27.75	27.52	27.19	-0.23 (1.31) [0.864]	-0.56 (1.29) [0.664]	0.33 (1.65) [0.839]
Payments	58.41	51.53	57.07	-6.89 (3.62) [0.057]	-1.34 (3.95) [0.734]	-5.55 (4.73) [0.241]
Statement Balance	278.41	282.85	283.45	4.44 (8.81) [0.614]	5.04 (8.52) [0.554]	-0.60 (11.05) [0.957]
Spending	65.25	62.05	62.15	-3.20 (4.59) [0.486]	-3.10 (4.28) [0.468]	-0.10 (5.59) [0.986]
Spending (%)	27.32	28.59	27.99	1.27 (1.34) [0.346]	0.66 (1.32) [0.616]	0.61 (1.70) [0.721]
Revolving Debt	230.91	239.63	240.26	8.72 (8.59) [0.310]	9.35 (8.25) [0.257]	-0.63 (10.74) [0.953]
Credit Limit	366.11	362.14	369.84	-3.97 (9.86) [0.687]	3.73 (9.76) [0.702]	-7.70 (12.47) [0.537]
Miss Pay (%)	35.97	39.83	37.91	3.86* (1.69) [0.023]	1.94 (1.67) [0.246]	1.92 (2.14) [0.371]
Pay Min (%)	15.62	10.56	10.46	-5.06*** (1.11) [0.000]	-5.16*** (1.10) [0.000]	0.10 (1.35) [0.940]
Pay 50% (%)	1.85	2.71	0.10	0.86 (0.55) [0.116]	-1.76*** (0.23) [0.000]	2.62*** (0.52) [0.000]
Pay Full (%)	18.24	16.96	17.37	-1.28 (1.31) [0.328]	-0.87 (1.31) [0.509]	-0.41 (1.66) [0.803]
Pay Current Balance (%)	4.06	3.78	3.74	-0.28 (0.67) [0.678]	-0.31 (0.66) [0.636]	0.04 (0.84) [0.965]

Dynamic Effects on Payments (% Statement Balance) After 3 & 6 Cycles

Comparison	Dynamic (δ_3^k)					Dynamic (δ_6^k)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1 (vs C)	-0.21 (1.33)	1.47 (1.39)	0.22 (1.22)	-1.55 (1.57)	-7.09*** (1.26)	0.01 (1.30)	-1.26 (1.23)	-2.83 (1.62)
T2 (vs C)	-0.78 (1.31)	1.00 (1.37)	0.90 (1.18)	1.68 (1.44)	-7.42*** (1.24)	-0.38 (1.29)	-0.54 (1.19)	0.10 (1.56)
T1 (vs T2)	0.57 (1.75)	0.47 (1.75)	-0.69 (1.52)	-3.23 (1.90)	0.33 (1.64)	0.40 (1.64)	-0.72 (1.53)	-2.93 (2.01)
Time F.E.		X	X	X		X	X	X
Consumer Controls			X				X	
Consumer F.E.				X				X

With Consumer F.E.: Payments



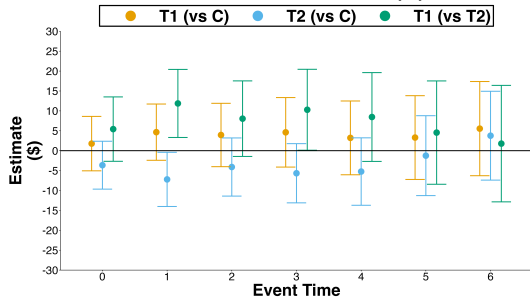
Effects On Payments (% Statement Balance) Pooled Across Cycles 0 to 6

Comparison	Pooled (δ^k)			
	(1)	(2)	(3)	(4)
T1 (vs C)	0.49 (1.06)	0.55 (1.06)	-0.67 (0.81)	-2.39 (1.23)
T2 (vs C)	0.35 (1.05)	0.37 (1.05)	0.29 (0.80)	1.04 (1.16)
T1 (vs T2)	0.14 (1.35)	0.18 (1.35)	-0.96 (1.02)	-3.44* (1.51)
Time F.E.		X	X	X
Consumer Controls			X	
Consumer F.E.				X

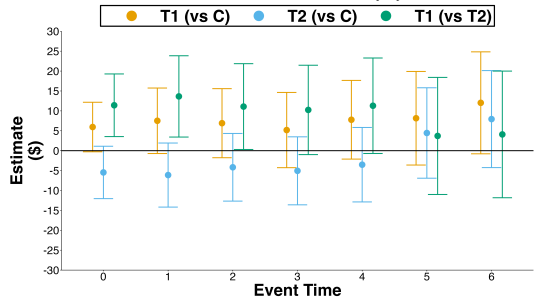
(percentage points, C mean: 34.6%)

With Consumer F.E.: Statement Balance (\$) & Revolving Debt (\$) 🚀

A. Statement Balance (\$)



B. Revolving Debt (\$)



Effects on Secondary Outcomes After 3 Cycles

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)
Payments	1.33 (3.37)	-3.48 (3.08)	4.81 (4.08)
Statement Balance	5.10 (3.65)	-4.48 (3.38)	9.57* (4.48)
Spending	-2.69 (3.68)	-4.57 (3.60)	1.88 (4.59)
Spending (%)	0.09 (1.20)	0.39 (1.19)	-0.31 (1.51)
Revolving Debt	3.79 (4.03)	-3.25 (3.70)	7.03 (4.90)
Credit Limit	0.39 (2.20)	-0.50 (2.05)	0.89 (2.75)
Miss Pay (%)	1.42 (1.47)	0.45 (1.44)	0.98 (1.85)
Pay Min (%)	-4.17*** (1.04)	-3.93*** (1.06)	-0.24 (1.3)
Pay 50% (%)	0.71 (0.60)	-2.88*** (0.26)	3.59*** (0.56)
Pay Full (%)	0.02 (1.31)	1.66 (1.26)	-1.64 (1.63)
Pay Current Balance (%)	-0.32 (0.78)	0.60 (0.80)	-0.92 (1.01)
Time F.E.	X	X	X
Consumer Controls	X	X	X

Robustness Effects on Secondary Outcomes After 3 Cycles

Outcome	T1 (vs C)			T2 (vs C)			T1 (vs T2)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Payments	-0.07 (3.84)	4.03 (4.00)	-3.72 (4.32)	-7.50* (3.44)	-3.09 (3.61)	-2.87 (3.81)	7.42 (4.81)	7.12 (4.82)	-0.85 (5.10)
Statement Balance	-3.34 (7.84)	2.45 (7.89)	4.61 (4.45)	-11.67 (7.55)	-6.00 (7.63)	-5.67 (3.79)	8.33 (9.89)	8.45 (9.87)	10.28* (5.19)
Spending	1.07 (4.35)	1.07 (4.53)	-0.42 (4.80)	-5.07 (4.06)	-4.77 (4.27)	-8.11 (4.92)	6.14 (5.57)	5.84 (5.58)	7.69 (6.14)
Spending (%)	2.46 (1.37)	1.93 (1.43)	-0.01 (1.60)	0.97 (1.36)	0.42 (1.42)	0.26 (1.58)	1.49 (1.82)	1.51 (1.81)	-0.27 (2.02)
Revolving Debt	-1.39 (7.58)	0.23 (7.65)	5.18 (4.82)	-6.69 (7.36)	-5.17 (7.46)	-5.06 (4.36)	5.30 (9.65)	5.39 (9.62)	10.24 (5.73)
Credit Limit	-5.61 (8.72)	-2.84 (8.68)	2.18 (2.68)	-3.28 (8.61)	-0.26 (8.56)	-0.84 (2.45)	-2.34 (11.02)	-2.58 (10.94)	3.02 (3.31)
Miss Pay (%)	2.79 (1.46)	0.84 (1.53)	3.24 (1.78)	2.30 (1.44)	0.38 (1.51)	1.29 (1.69)	0.49 (1.96)	0.46 (1.94)	1.95 (2.20)
Pay Min (%)	-4.76*** (1.05)	-4.70*** (1.12)	-4.62*** (1.42)	-4.77*** (1.04)	-4.69*** (1.11)	-3.84** (1.46)	0.01 (1.38)	-0.01 (1.37)	-0.78 (1.80)
Pay 50% (%)	0.75 (0.56)	0.68 (0.60)	1.36 (0.8)	-2.59*** (0.12)	-2.74*** (0.25)	-3.23*** (0.71)	3.35*** (0.55)	3.43*** (0.56)	4.59*** (0.95)
Pay Full (%)	0.36 (1.39)	1.33 (1.45)	-1.45 (1.77)	0.42 (1.38)	1.57 (1.44)	3.22* (1.60)	-0.06 (1.84)	-0.24 (1.84)	-4.66* (2.13)
Pay Current Balance (%)	-0.65 (0.74)	-0.08 (0.80)	-1.83 (1.23)	-0.01 (0.77)	0.55 (0.83)	0.18 (1.19)	-0.64 (1.03)	-0.63 (1.04)	-2.01 (1.55)
Time F.E.		X	X		X	X		X	X
Consumer F.E.			X			X			X

Effects on Secondary Outcomes After 6 Cycles

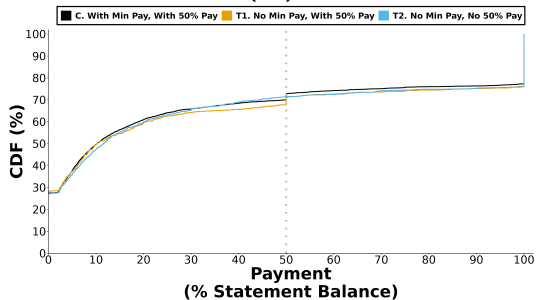
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-9.31** (3.32)	-14.08*** (4.45)	-1.95 (3.65)	-1.19 (4.44)	-7.36 (4.38)	-12.89* (5.62)
Statement Balance	6.03 (4.96)	5.54 (6.04)	3.92 (4.94)	3.77 (5.69)	2.11 (6.33)	1.78 (7.46)
Spending	-6.83 (4.27)	-4.28 (5.48)	-3.27 (3.99)	-6.73 (5.51)	-3.56 (5.27)	2.45 (7.00)
Spending (%)	-0.41 (1.28)	-0.26 (1.75)	0.71 (1.22)	0.52 (1.64)	-1.12 (1.6)	-0.78 (2.16)
Revolving Debt	10.76* (5.32)	12.02 (6.53)	8.79 (5.14)	7.94 (6.22)	1.97 (6.68)	4.09 (8.12)
Credit Limit	-2.15 (4.33)	-0.14 (4.97)	0.35 (4.17)	0.08 (4.74)	-2.50 (5.47)	-0.22 (6.22)
Miss Pay (%)	4.39** (1.63)	5.70*** (1.91)	1.92 (1.61)	2.71 (1.88)	2.47 (2.06)	2.99 (2.41)
Pay Min (%)	-4.60*** (1.09)	-4.82*** (1.56)	-4.34*** (1.06)	-3.99** (1.50)	-0.26 (1.32)	-0.84 (1.93)
Pay 50% (%)	0.94 (0.54)	1.60* (0.80)	-1.93*** (0.24)	-2.20*** (0.71)	2.87*** (0.52)	3.80*** (0.95)
Pay Full (%)	-2.44 (1.25)	-3.80* (1.78)	-0.72 (1.23)	0.66 (1.70)	-1.72 (1.57)	-4.46* (2.21)
Pay Current Balance (%)	-0.48 (0.66)	-1.96 (1.17)	-0.33 (0.66)	-0.79 (1.17)	-0.16 (0.84)	-1.18 (1.49)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Pooled Effects On Secondary Outcomes

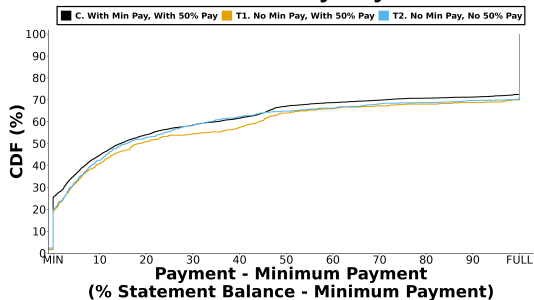
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments	-3.83 (2.19)	-8.85* (3.43)	-3.31 (2.13)	-2.73 (3.12)	-0.52 (2.74)	-6.12 (4.15)
Statement Balance	4.39 (2.42)	3.85 (3.63)	-2.29 (2.35)	-3.40 (3.08)	6.68 (3.04)	7.25 (4.21)
Spending	-2.49 (2.13)	-0.25 (3.93)	-2.32 (2.10)	-5.87 (4.11)	-0.17 (2.70)	5.61 (5.12)
Spending (%)	-0.38 (0.71)	-0.46 (1.29)	0.72 (0.71)	0.56 (1.25)	-1.10 (0.9)	-1.02 (1.61)
Revolving Debt	6.29* (2.78)	7.61 (3.95)	-0.10 (2.72)	-1.81 (3.64)	6.39 (3.50)	9.42* (4.77)
Credit Limit	-0.89 (1.72)	0.91 (2.26)	-0.09 (1.64)	-0.41 (2.11)	-0.80 (2.17)	1.32 (2.80)
Miss Pay (%)	1.72 (0.90)	3.43* (1.34)	1.29 (0.90)	2.18 (1.30)	0.43 (1.15)	1.26 (1.66)
Pay Min (%)	-3.53*** (0.65)	-3.94*** (1.17)	-3.38*** (0.64)	-3.29** (1.18)	-0.15 (0.80)	-0.65 (1.49)
Pay 50% (%)	0.70* (0.29)	1.35* (0.62)	-2.46*** (0.15)	-2.80*** (0.67)	3.16*** (0.28)	4.15*** (0.81)
Pay Full (%)	-1.00 (0.85)	-2.43 (1.42)	1.20 (0.82)	2.75* (1.30)	-2.20* (1.07)	-5.18*** (1.72)
Pay Current Balance (%)	-0.67 (0.40)	-2.17* (1.01)	0.26 (0.42)	-0.16 (0.97)	-0.93 (0.52)	-2.01 (1.26)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Distribution of Payments, After Three Cycles 🚀

A. CDF Payments (%)

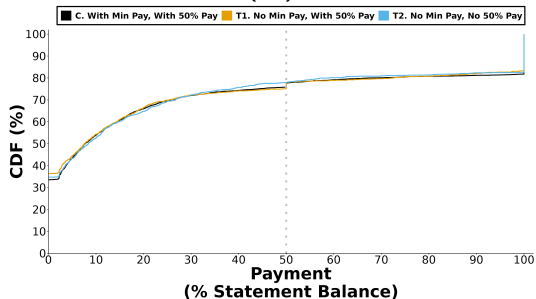


B. CDF Excess Payments (\$), Conditional On Any Payment

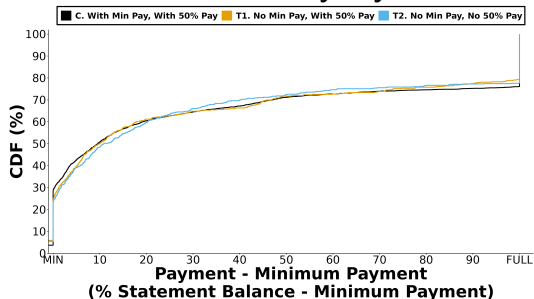


Distribution of Payments, After Six Cycles

A. CDF Payments (%)

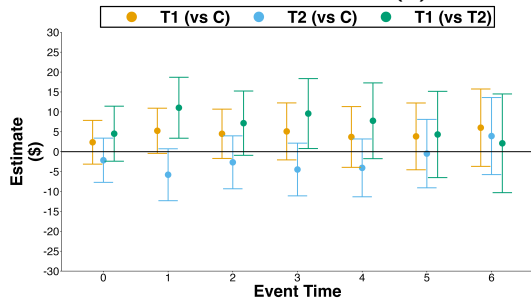


B. CDF Excess Payments (\$), Conditional On Any Payment

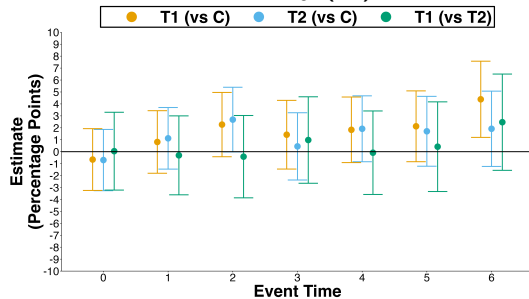


Statement Balance (\$), Miss Pay (%) 🚀

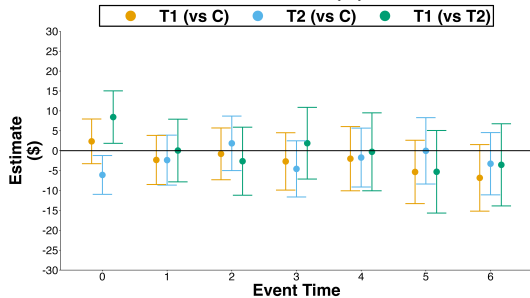
A. Statement Balance (\$)



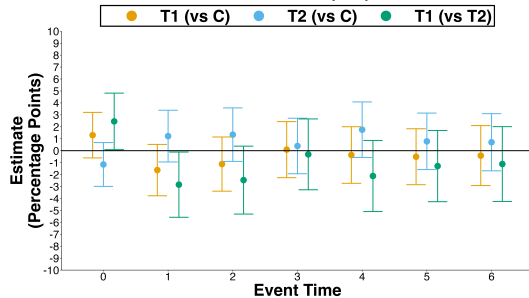
B. Miss Pay (%)



A. Spending (\$)

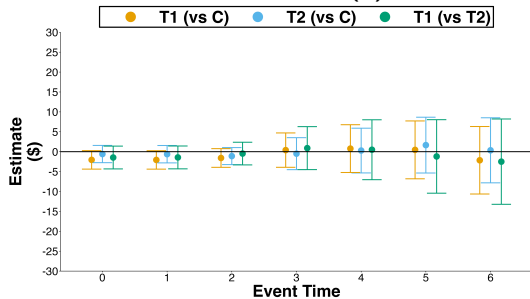


B. Spending (%)

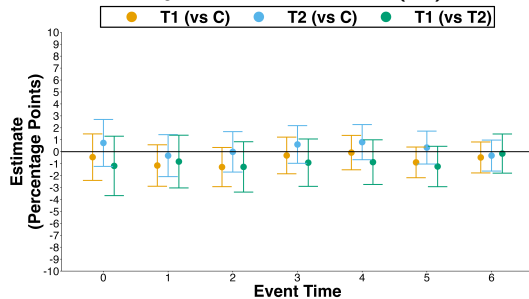


Credit Limit (\$), Pay Current Balance (%) 🚀

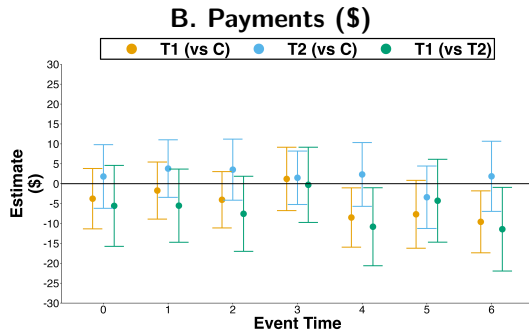
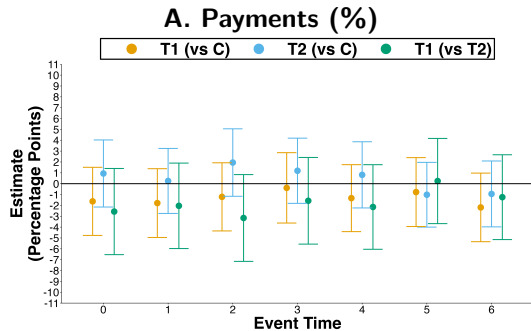
A. Credit Limit (\$)



B. Pay Current Balance (%)



No Autopay Subsample: Payments 🚀



No Autopay Subsample: Dynamic Effects After 6 Cycles

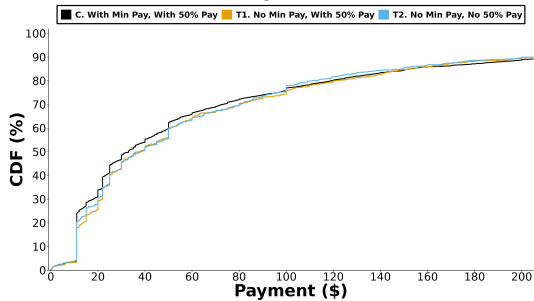
Outcome	T1 (vs C)		T2 (vs C)		T1 (vs T2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Payments (%)	-2.18 (1.61)	-2.97 (2.08)	-0.94 (1.55)	-0.24 (2.02)	-1.24 (1.99)	-2.73 (2.58)
Payments	-9.58* (3.97)	-10.48* (5.08)	1.85 (4.50)	3.84 (5.37)	-11.44* (5.36)	-14.31* (6.51)
Statement Balance	5.66 (5.8)	3.30 (6.93)	8.07 (5.75)	8.87 (6.78)	-2.41 (7.42)	-5.57 (8.82)
Spending	-9.12 (5.10)	-7.76 (6.94)	1.15 (5.00)	0.10 (6.69)	-10.27 (6.48)	-7.86 (8.65)
Spending (%)	-1.12 (1.73)	-0.42 (2.37)	0.03 (1.58)	-0.39 (2.15)	-1.15 (2.11)	-0.03 (2.89)
Revolving Debt	10.31 (6.09)	9.85 (7.46)	9.13 (6.04)	7.11 (7.43)	1.18 (7.76)	2.74 (9.47)
Credit Limit	-3.17 (5.2)	-2.24 (5.78)	5.93 (5.20)	6.46 (5.89)	-9.09 (6.73)	-8.70 (7.50)
Miss Pay (%)	5.03* (2.21)	5.38* (2.55)	2.01 (2.12)	2.77 (2.42)	3.02 (2.75)	2.61 (3.16)
Pay Min (%)	-4.99*** (1.24)	-5.17*** (1.93)	-5.63*** (1.18)	-6.05*** (1.77)	0.64 (1.46)	0.88 (2.33)
Pay 50% (%)	0.81 (0.81)	2.22 (1.15)	-2.66*** (0.38)	-3.05*** (1.08)	3.47*** (0.78)	5.26*** (1.41)
Pay Full (%)	-3.84* (1.62)	-3.81 (2.35)	-1.33 (1.58)	0.43 (2.22)	-2.51 (2.01)	-4.24 (2.89)
Pay Current Balance (%)	-0.29 (0.92)	-0.96 (1.53)	-0.45 (0.88)	-0.42 (1.52)	0.16 (1.13)	-0.54 (1.94)
Time F.E.	X	X	X	X	X	X
Consumer Controls	X		X		X	
Consumer F.E.		X		X		X

Effects On Portfolio of Credit Card Statement Balances 🍷

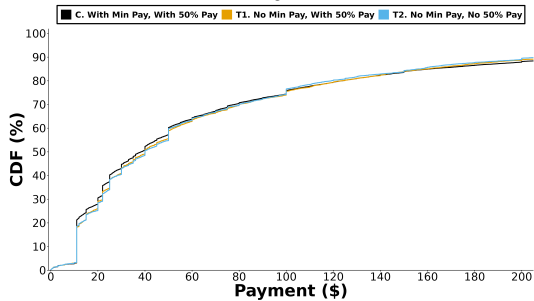
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Portfolio Statement Balances	36.55 (51.85)	6.60 (53.41)	29.94 (66.98)	\$3,681
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Distribution of Payments (\$) 🚀

A. Cycle 2



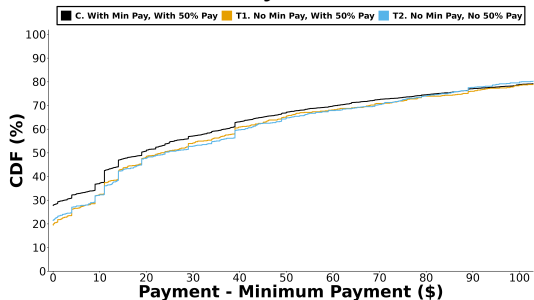
B. Pooled Cycles 0 to 6



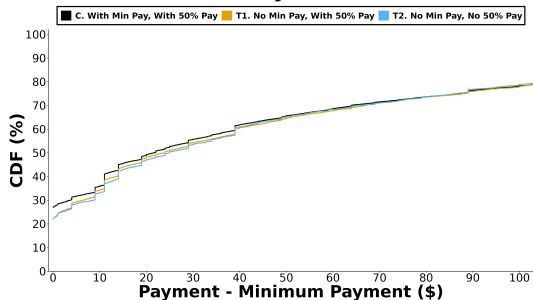
Distribution of Excess Payments (\$)



A. Cycle 2

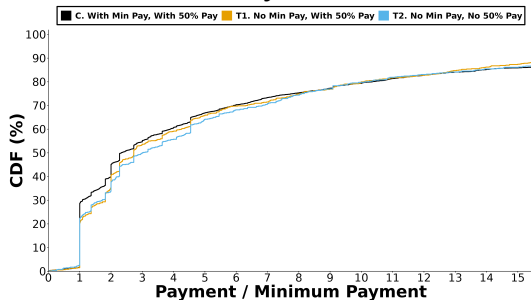


B. Pooled Cycles 0 to 6

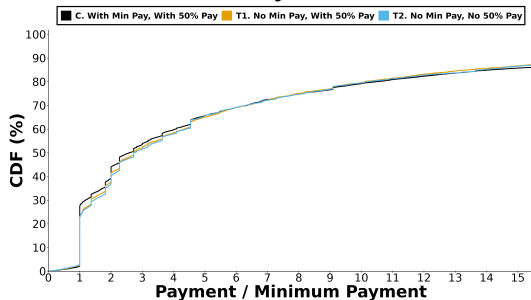


Distribution of Ratio of Payments To Minimum Payment

A. Cycle 2



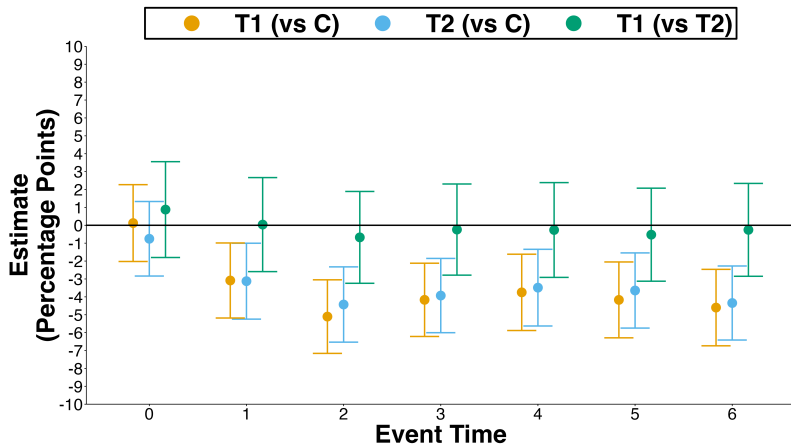
B. Pooled Cycles 0 to 6



Effects On Heuristic Payment Amounts

Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Pay Multiples Of Minimum	−0.76*** (0.25)	−0.94*** (0.23)	0.18 (0.29)	3.48%
Pay Round Number	2.51*** (0.62)	3.61*** (0.62)	−1.10 (0.80)	9.58%
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

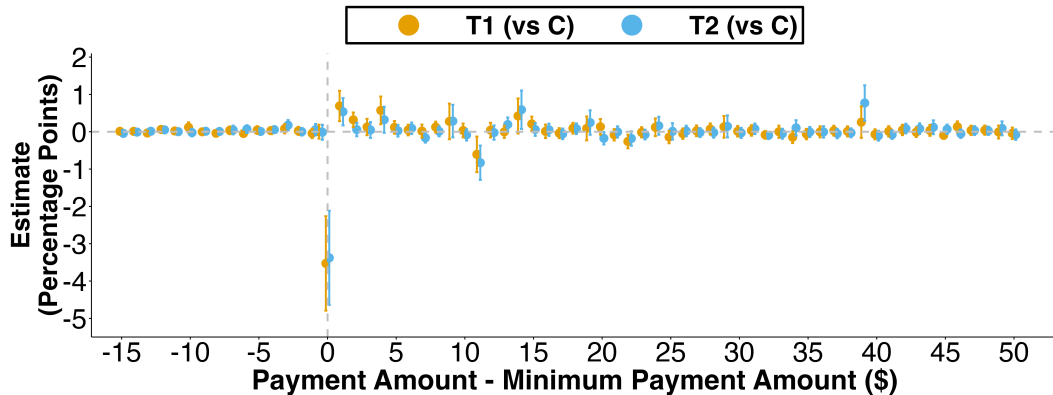
Effects On Payment Distribution: 29% Less Likely To Only Pay The Minimum



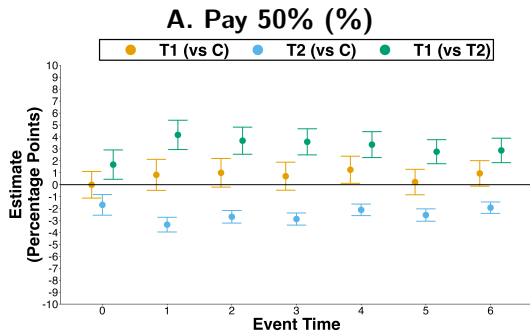
δ_6^k (percentage points, C mean: 15.6%):

T1 -4.60^{***} (s.e. 1.09) **T2** -4.34^{***} (s.e. 1.06) **T1 (vs T2)** -0.26 (s.e. 1.32)

Effects On Payments Proximate To Minimum Payment Amount

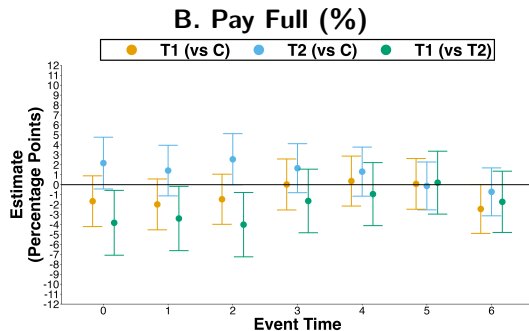


50% Payment Option Makes Consumers Initially Less Likely To Pay In Full



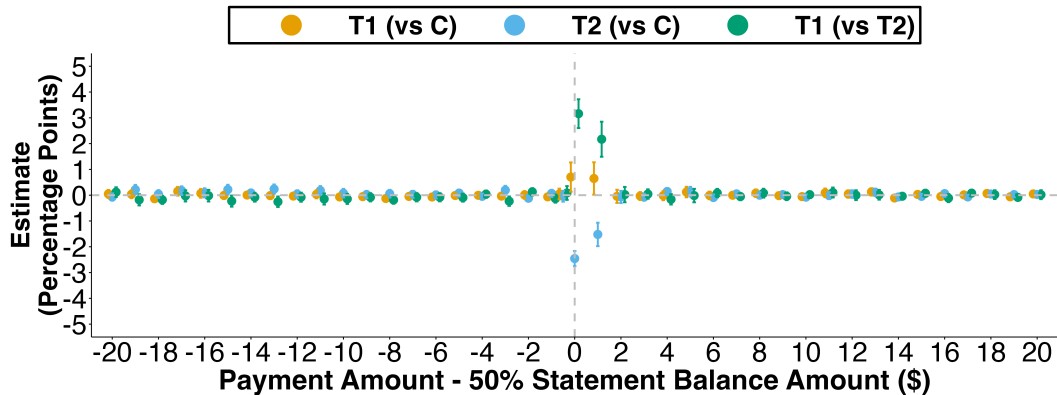
δ_6^k (percentage points):

T1 (vs C) 0.94 (s.e. 0.54)
T2 (vs C) -1.93*** (s.e. 0.24)
T1 (vs T2) 2.87*** (s.e. 0.52)
(C mean: 1.9%)

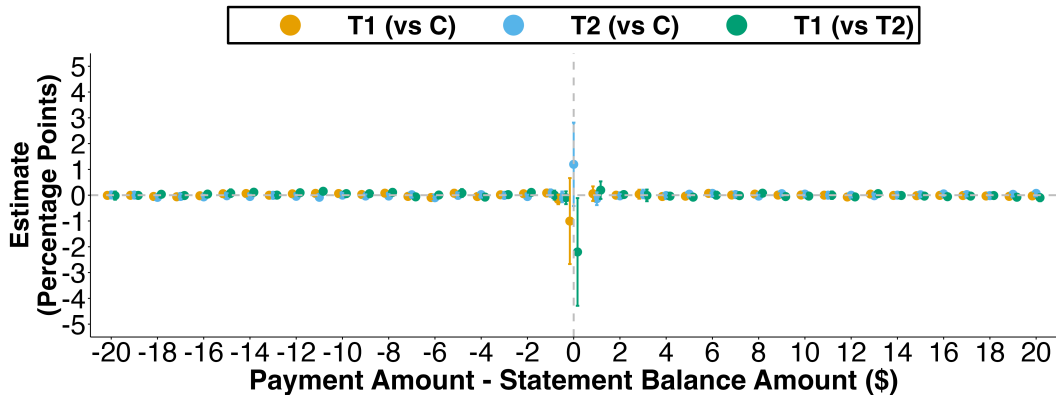


T1 (vs C) -2.44 (s.e. 1.25)
T2 (vs C) -0.72 (s.e. 1.23)
T1 (vs T2) -1.72 (s.e. 1.57)
(C mean: 18.2%)

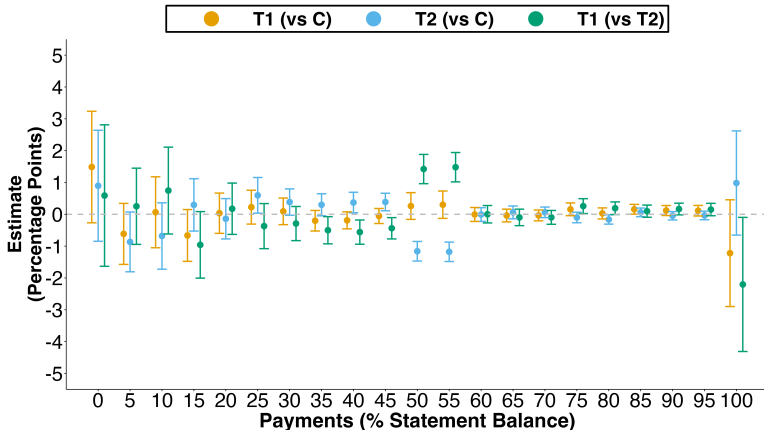
Effects On Payments Proximate To 50% Statement Balance Amount 🚀



Effects On Payments Proximate To Statement Balance Amount 🚀



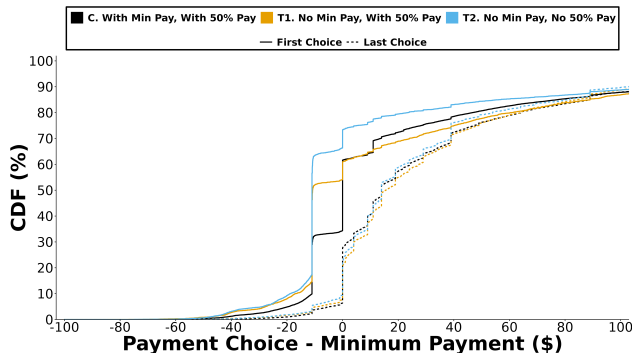
Limited Effects On The Distribution Of Payments (% Statement Balance)



T1 (vs T2): $\uparrow Pr(\text{Payment} \in \{(45\%, 55\%]\})$, $\downarrow Pr(\text{Payment} \in \{(30\%, 45\%], 100\%\})$

Shrouding shifts payments towards small, round amounts, away from multiples of the minimum

CDFs Of Initial Vs. Last Payment Choices Within Payment Cycle



Pr (Initial Payment Choice < Minimum Payment):

T1 (vs C) 19.44*** (s.e. 1.16) T2 (vs C) 30.50*** (s.e. 1.02)

T1 (vs T2) -11.06*** (s.e. 1.38) (percentage points, C mean: 33.7%)

A. Initial Payment Choices

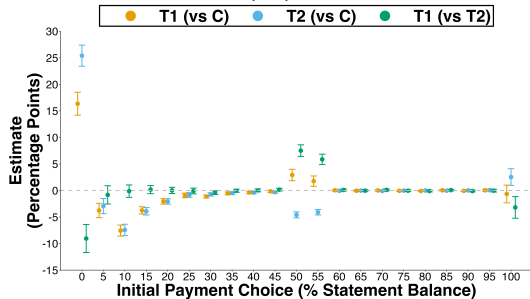
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Initial Payment Choice (% Balance)	-0.90 (0.87)	-4.33*** (0.81)	3.43*** (1.07)	26.83%
Pr (Initial Payment Choice < Minimum)	19.44*** (1.16)	30.50*** (1.02)	-11.06*** (1.38)	33.74%
Number of Payment Choices	0.40*** (0.11)	0.85*** (0.11)	-0.46*** (0.14)	3.82
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

B. Initial Payment Choices, Excluding Blanks & Zeros

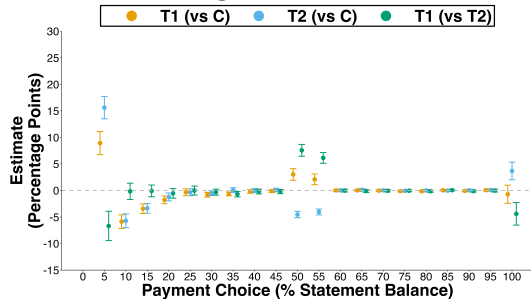
Outcome	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean
Initial Payment Choice (% Balance)	-0.22 (0.86)	-1.82* (0.83)	1.60 (1.07)	30.19%
Pr (Initial Payment Choice < Minimum)	15.59*** (1.08)	23.57*** (1.04)	-7.98*** (1.38)	21.73%
Number of Payment Choices	0.10 (0.11)	0.35*** (0.11)	-0.25* (0.14)	3.01
Time F.E.	X	X	X	
Consumer Controls	X	X	X	

Effects On Initial Payment Choice 🚀

A. Initial Payment Choices (%)

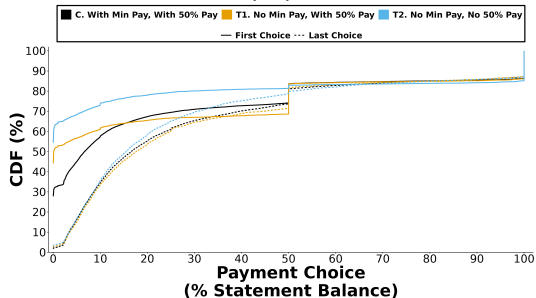


B. Initial Payment Choices (%), Excluding Blanks & Zeros

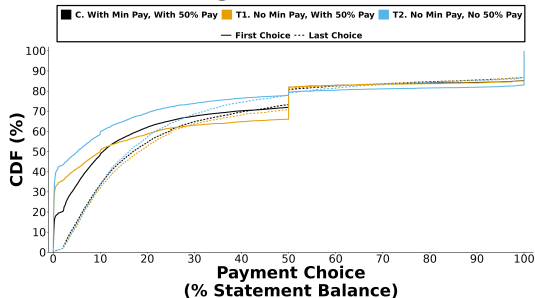


Initial Vs. Last Payment Choices Within Payment Cycle 🚀

A. CDF Initial Payment Choices (%)



B. CDF Initial Payment Choices (%), Excluding Blanks & Zeros



Empirical Support For New Macroeconomic Modeling Approach

- Heterogeneous agent models (“HANK”) include hand-to-mouth consumers with zero liquid wealth **experiencing binding borrowing constraints** (Kaplan & Violante, 2022)
- Lee & Maxted (2026): **binding borrowing constraints inconsistent with SCF data**
 - Consumers often borrow intermediate levels of credit card debt
 - This result could be driven by under-reporting of credit card debt in SCF (Zinman, 2009)
- I also find cardholders with low liquid cash are not borrowing constrained
- My empirical support for Lee & Maxted (2026)’s theory has broad implications
 - Affects how households respond to fiscal and monetary policy

Follow Keys & Wang (2019) Defining Near Minimum Payers 🗝

Anchoring predicts largest results for unconstrained cardholders “near minimum payers”:

$$\text{payments} \in \{(Minimum, Minimum + \$50] \setminus Full\}$$

18% my data, 20% in Keys & Wang (2019)

A. Payments (%)

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Near-Minimum	2.89 (1.73)	-0.56 (1.54)	3.45 (2.07)	22.18%	1,226
Not-Near-Minimum	-1.44 (0.92)	0.28 (0.92)	-1.72 (1.17)	37.42%	5,488
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

Follow Keys & Wang (2019) Defining Near Minimum Payers 🗝

Anchoring predicts largest results for unconstrained cardholders “near minimum payers”:

$$\text{payments} \in \{(Minimum, Minimum + \$50] \setminus Full\}$$

18% my data, 20% in Keys & Wang (2019)

B. Payments (\$)

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Near-Minimum	0.43 (2.56)	-2.41 (2.39)	2.84 (3.00)	\$36.76	1,226
Not-Near-Minimum	-4.05 (2.56)	-3.79 (2.53)	-0.26 (3.23)	\$69.24	5,488
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

HTE: Payments (%) By Pre-Experiment Credit Card Utilization

Portfolio utilization: 25% (median), 36% (mean)
16% cardholders with 90%+ portfolio utilization (borrowing constrained)

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Q1 Card Utilization (< 60%)	-1.89 (1.65)	3.70* (1.60)	-5.59** (2.08)	52.23%	2,238
Q2 Card Utilization (60–91%)	1.57 (1.40)	-1.19 (1.31)	2.76 (1.72)	30.08%	2,238
Q3 Card Utilization (91%+)	-1.45 (1.15)	-1.30 (1.17)	-0.15 (1.44)	21.38%	2,238
Q1 Portfolio Utilization (< 12%)	-1.68 (1.43)	0.37 (1.47)	-2.05 (1.86)	33.24%	2,238
Q2 Portfolio Utilization (12–45%)	-2.31 (1.43)	0.94 (1.40)	-3.25 (1.83)	33.72%	2,238
Q3 Portfolio Utilization (45%+)	1.66 (1.45)	-0.96 (1.36)	2.62 (1.78)	36.85%	2,238
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

HTE: Payments (\$) By Pre-Experiment Credit Card Utilization

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Q1 Card Utilization (< 60%)	−8.30* (4.15)	−1.53 (4.14)	−6.78 (5.30)	\$80.49	2,238
Q2 Card Utilization (60–91%)	0.01 (3.31)	−5.73 (2.99)	5.74 (3.90)	\$58.93	2,238
Q3 Card Utilization (91%+)	−2.06 (3.12)	−0.13 (3.33)	−1.93 (4.02)	\$50.19	2,238
Q1 Portfolio Utilization (< 12%)	−2.93 (3.83)	2.13 (4.07)	−5.06 (5.00)	\$64.13	2,238
Q2 Portfolio Utilization (12–45%)	−4.57 (3.26)	−1.15 (3.24)	−3.43 (4.26)	\$60.13	2,238
Q3 Portfolio Utilization (45%+)	−0.96 (3.95)	−7.82* (3.61)	6.86 (4.81)	\$65.48	2,238
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

HTE: Payments (%) By Pre-Experiment Liquid Cash Balances

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Liquid Cash < \$1	1.32 (2.28)	1.01 (2.22)	0.31 (2.97)	25.15%	1,025
Liquid Cash \$1+	-1.36 (0.88)	0.20 (0.86)	-1.56 (1.10)	36.33%	5,689
Liquid Cash < \$501	-0.46 (1.06)	0.70 (1.06)	-1.16 (1.34)	31.37%	3,796
Liquid Cash \$501+	-0.80 (1.3)	-0.45 (1.23)	-0.35 (1.60)	38.78%	2,918
Liquid Cash < \$1,001	-1.11 (0.95)	0.42 (0.95)	-1.53 (1.21)	32.11%	4,726
Liquid Cash \$1,001+	0.25 (1.58)	-0.22 (1.50)	0.47 (1.96)	40.60%	1,988
Minimum Liquid Cash < \$1	-1.47 (1.02)	0.68 (1.03)	-2.15 (1.31)	32.10%	4,230
Minimum Liquid Cash \$1+	-0.25 (1.37)	-0.15 (1.33)	-0.09 (1.70)	38.99%	2,484
Minimum Liquid Cash < \$501	-0.88 (0.83)	0.61 (0.83)	-1.49 (1.06)	33.64%	6,252
Minimum Liquid Cash \$501+	1.50 (3.33)	0.43 (3.34)	1.07 (4.07)	48.04%	462
Minimum Liquid Cash < \$1,001	-0.95 (0.82)	0.21 (0.82)	-1.16 (1.04)	33.94%	6,422
Minimum Liquid Cash \$1,001+	1.77 (4.46)	5.87 (4.55)	-4.10 (5.77)	49.24%	292
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

HTE: Payments (\$) By Pre-Experiment Liquid Cash Balances

Sample	T1 (vs C) (1)	T2 (vs C) (2)	T1 (vs T2) (3)	Control Mean	Consumers
Liquid Cash < \$1	-2.54 (4.48)	4.58 (5.38)	-7.13 (6.58)	\$43.44	1,025
Liquid Cash \$1+	-5.17* (2.42)	-4.92* (2.29)	-0.24 (2.99)	\$66.85	5,689
Liquid Cash < \$501	-3.44 (2.69)	-0.67 (2.76)	-2.77 (3.47)	\$58.73	3,796
Liquid Cash \$501+	-3.28 (3.51)	-5.15 (3.30)	1.86 (4.30)	\$69.07	2,918
Liquid Cash < \$1,001	-3.55 (2.49)	-1.94 (2.49)	-1.61 (3.18)	\$60.45	4,726
Liquid Cash \$1,001+	-1.86 (4.24)	-4.53 (3.90)	2.67 (5.10)	\$69.94	1,988
Minimum Liquid Cash < \$1	-4.37 (2.54)	-1.92 (2.67)	-2.45 (3.32)	\$59.70	4,230
Minimum Liquid Cash \$1+	-3.69 (4.03)	-6.22 (3.49)	2.53 (4.78)	\$69.44	2,484
Minimum Liquid Cash < \$501	-2.66 (2.26)	-2.59 (2.18)	-0.06 (2.84)	\$62.05	6,252
Minimum Liquid Cash \$501+	-15.50 (8.53)	1.52 (8.83)	-17.02 (9.97)	\$79.79	462
Minimum Liquid Cash < \$1,001	-3.49 (2.22)	-3.34 (2.16)	-0.16 (2.79)	\$62.41	6,422
Minimum Liquid Cash \$1,001+	-18.52 (10.03)	3.25 (10.85)	-21.77 (12.49)	\$81.69	292
Time F.E.	X	X	X		
Consumer Controls	X	X	X		

Distinguishing Anchoring from Targeting

Anchors: *Irrelevant* numbers influence choices e.g., thinking of your (randomly-assigned) last two digits of SSN influences choices (Tversky & Kahneman, 1974; Ariely et al., 2003)

Targets (reference-dependent preferences): Consumers exert effort to meet/exceed *relevant* numbers e.g., marathon time goals (Heath, Larrick, & Wu, 1999; Allen et al., 2017)

Distinguishing Anchoring from Targeting

Anchors: *Irrelevant* numbers influence choices e.g., thinking of your (randomly-assigned) last two digits of SSN influences choices (Tversky & Kahneman, 1974; Ariely et al., 2003)

Targets (reference-dependent preferences): Consumers exert effort to meet/exceed *relevant* numbers e.g., marathon time goals (Heath, Larrick, & Wu, 1999; Allen et al., 2017)

Distinguish using **“VPA” Value Proximity Analysis** (Bartels, Herzog, & Sussman, 2024)

VPA Predicts:

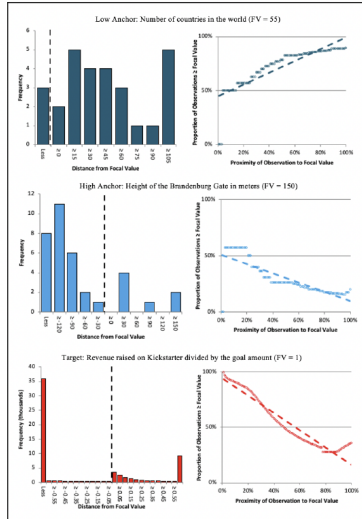
Anchoring: $\alpha \approx 50\%$ (symmetric), $\gamma > 0$ if low anchor ($\gamma < 0$ if high), $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} \neq 0$

Targeting: $\alpha \approx 100\%$ (asymmetric), $\gamma < 0$, $\beta^{T1}, \beta^{T2}, \theta^{T1}, \theta^{T2} = 0$

$$\Pr(p_{g,w} \geq V) = \alpha + \beta_g^{T1} T1_w + \beta_g^{T2} T2_w \\ + \gamma Proximity_w + \theta^{T1} (Proximity_w \times T1_g) + \theta^{T2} (Proximity_w \times T2_g) + \varepsilon_{g,w}$$

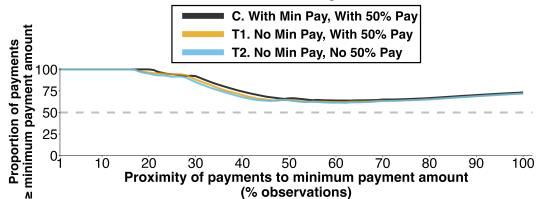
Example Histogram & VPA Charts (Bartels, Herzog, & Sussman, 2024) 🚀

Example Histograms and VPA Charts

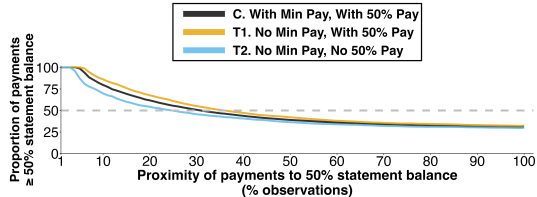


Value Proximity Analysis (VPA): Payments Proximate To Focal Amounts

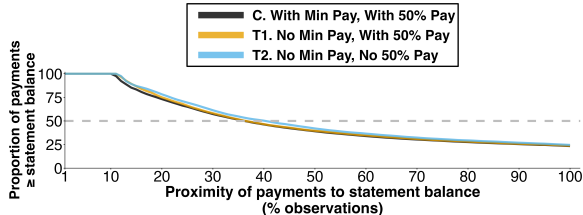
A. Minimum Payment



B. 50% Statement Balance



C. Statement Balance



VPA Results Inconsistent With Anchoring

	Minimum Payment	50% Statement Balance	Statement Balance
	(1)	(2)	(3)
Intercept	98.510*** (0.383)	76.831*** (0.940)	89.091*** (0.822)
T1	-1.403 (0.952)	5.290** (2.034)	1.380 (1.875)
T2	-2.840*** (0.995)	-7.568*** (2.410)	3.377 (1.767)
Proximity	-0.406*** (0.006)	-0.584*** (0.009)	-0.798*** (0.005)
Proximity × T1	-0.005 (0.015)	-0.044* (0.018)	-0.012 (0.011)
Proximity × T2	0.008 (0.015)	0.078*** (0.023)	-0.016 (0.010)
T2 – T1: Intercept	-1.438 (1.253)	-12.858*** (2.862)	1.997 (2.298)
T2 – T1: Proximity	0.013 (0.019)	0.122*** (0.027)	-0.004 (0.013)